

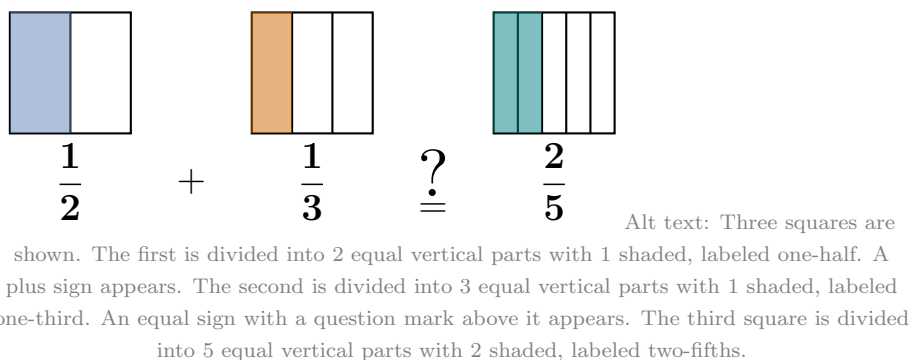
Fractions: Adding and Subtracting

A Common Mistake

A student is asked to add $\frac{1}{2}$ and $\frac{1}{3}$ and writes the following. Is their work correct?

$$\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$$

To answer this question, let's look at a visual representation of the student's logic.



Since $\frac{1}{3}$ is *added* to $\frac{1}{2}$, the result should be greater than $\frac{1}{2}$. However, you can see that the student's answer of $\frac{2}{5}$ is actually less than $\frac{1}{2}$. This shows that simply adding the numerators and denominators doesn't work! To get the correct answer, we need to divide each whole into the same number of pieces.

Fractions can only be added or subtracted when the pieces are the same size.

Recall that the denominator of a fraction tells you how many equal-sized pieces a whole has been divided into. If we decide to divide each whole into six equal pieces, the original fractions can be rewritten as equivalent fractions with a denominator of 6 by multiplying by 1 in the appropriate form. Below, the fraction $\frac{1}{2}$ has been multiplied by 1 in form $\frac{3}{3}$ to rewrite it as an equivalent

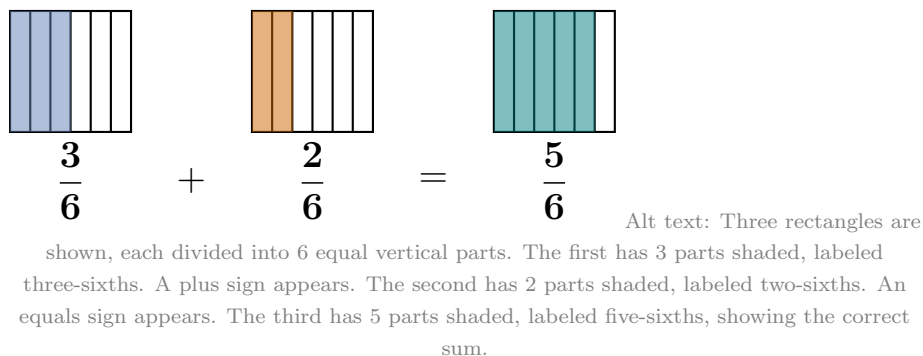
Fractions: Adding and Subtracting

fraction with a denominator of 6. Multiply the fraction $\frac{1}{3}$ by 1 in the appropriate form to rewrite it as an equivalent fraction with a denominator of 6.

$$\begin{aligned}\frac{1}{2} &= \frac{1}{2} \cdot \frac{3}{3} \\ &= \frac{3}{6}\end{aligned}$$

$$\begin{aligned}\frac{1}{3} &= \frac{1}{3} \cdot \frac{2}{2} \\ &= \frac{2}{6}\end{aligned}$$

Now we can visually represent the sum $\frac{1}{2} + \frac{1}{3}$ as an equivalent sum with equal pieces:

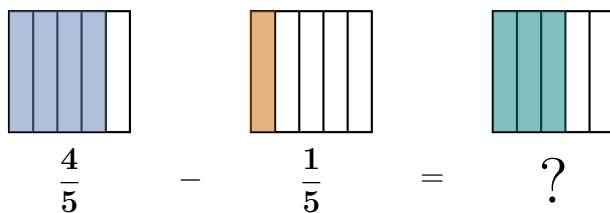


Combining 3 sixths and 2 sixths results in a total of 5 sixths. The entire problem is written mathematically below:

$$\begin{aligned}\frac{1}{2} + \frac{1}{3} &= \frac{1}{2} \cdot \frac{3}{3} + \frac{1}{3} \cdot \frac{2}{2} \\ &= \frac{3}{6} + \frac{2}{6} \\ &= \frac{5}{6}\end{aligned}$$

Remark 1. To learn mathematics, your solutions should be written using clear, logical steps as shown above; this process will deepen your understanding, prevent errors and serve as a test of your knowledge—if you cannot clearly communicate your work, you have not yet fully grasped the concept.

Problem 1 Use the visual representation to find the difference.



Alt text: A visual fraction subtraction problem shown using three identical rectangles divided vertically into five equal columns. The first rectangle has four columns shaded and is labeled four-fifths. This is followed by a minus sign. The second rectangle has one column shaded and is labeled one-fifth. This is followed by an equal sign. The third rectangle has three columns shaded and is labeled with a large question mark.

$$\frac{4}{5} - \frac{1}{5} = \frac{3}{5}$$

Procedure 1 (Adding and Subtracting Fractions). (a) **Find a Common Denominator**

Before adding or subtracting, the pieces must be the same size. If the denominators are different, rewrite the fractions as equivalent fractions with a common denominator. This is accomplished by multiplying each fraction by 1 in the appropriate form.

(b) **Add or Subtract the Numerators**

This tells you how many pieces you have.

(c) **Keep the Denominator the Same**

The size of the pieces doesn't change during addition or subtraction, so the denominator stays the same.

(d) **Simplify**

Check to see if the resulting fraction can be simplified. If so, simplify.

The Least Common Denominator

When adding or subtracting fractions with different denominators, we will use the *smallest* or *least common denominator*. This will make the simplification at the end as easy as possible.

Definition 1 (Multiples). The **multiples of a number** are found by multiplying the number by the counting numbers: $\{1, 2, 3, 4, \dots\}$. For example, the first three multiples of 4 are shown below.

$$4 \cdot 1 = 4$$

$$4 \cdot 2 = 8$$

$$4 \cdot 3 = 12$$

Problem 2 List the first five multiples of 3.

Multiple Choice:

(a) 0, 3, 6, 9, 12

(b) 3, 6, 9, 12, 15 ✓

(c) 6, 9, 12, 15, 18

(d) 3, 9, 12, 15, 18

Definition 2 (Least Common Denominator LCD). The **least common denominator** is the smallest number that is a multiple of two or more denominators. Equivalently, it is the smallest number that is evenly divisible by two or more denominators.

Procedure 2 (Finding the Least Common Denominator). Write the multiples of each denominator until you find the smallest one that appears in each list. **Alternatively**, list the multiples of only the largest denominator and stop when you find the smallest one that is evenly divisible by the other denominators.

Example 1. Find the LCD of $\frac{3}{4}$ and $\frac{1}{7}$.

Explanation. List the multiples of 4 and 7 as shown below.

Multiples of 4 : 4, 8, 12, 16, 20, 24, **28**, 32, 36, ...

Multiples of 7 : 7, 14, 21, **28**, 35, 42, 49, 56, 63, ...

Since 28 is the smallest number that appears in *both* lists, the LCD = 28.

Remark 2. Note that it is more efficient to list only the multiples of 7 and look for the smallest number in the list that is evenly divisible by 4. Had we done so, the list would've stopped at 28.

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Problem 3 Now that you know that the LCD of $\frac{3}{4}$ and $\frac{1}{7}$ is 28, rewrite each fraction as an equivalent fraction with a denominator of 28. Then add the fractions and simplify, if possible.

$$\frac{3}{4} = \frac{21}{28}$$

$$\frac{1}{7} = \frac{4}{28}$$

Hint: Multiply each fraction by 1 in the appropriate form. For example, to write $\frac{3}{4}$ as an equivalent fraction with a denominator of 28, multiply it by 1 in the form $\frac{7}{7}$.

$$\frac{3}{4} + \frac{1}{7} = \frac{25}{28}$$

Problem 4 Follow the steps below to find the sum $\frac{1}{4} + \frac{2}{3}$.

(a) Find the LCD of the fractions.

$$LCD = 12$$

Hint: To find the LCD, list the multiples of 4 and look for the smallest one that is evenly divisible by 3.

(b) Rewrite each fraction as an equivalent fraction with the least common denominator.

$$\frac{1}{4} = \frac{3}{12}$$

$$\frac{2}{3} = \frac{8}{12}$$

(c) Add the fractions. Simplify your answer, if possible.

$$\frac{1}{4} + \frac{2}{3} = \frac{11}{12}$$

Problem 5 Follow the steps below to find the difference $\frac{7}{10} - \frac{1}{6}$.

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- (a) Find the LCD of the fractions.

$$LCD = 30$$

- (b) Rewrite each fraction as an equivalent fraction with the least common denominator.

$$\frac{7}{10} = \frac{21}{30}$$

$$\frac{1}{6} = \frac{5}{30}$$

- (c) Subtract the fractions. Simplify your answer, if possible.

$$\frac{7}{10} - \frac{1}{6} = \frac{8}{15}$$

Problem 6 A community land trust is dividing a large lot to combat food insecurity. They decide to use $\frac{2}{3}$ of the lot for growing fresh vegetables to donate to local food pantries and $\frac{1}{5}$ of the lot for a fruit orchard. Follow the steps below to find the fraction of the land that is dedicated to food production.

- (a) Find the LCD of the fractions.

$$LCD = 15$$

Hint: To find the LCD, list the multiples of 5 and look for the smallest one that is evenly divisible by 3.

- (b) Rewrite each fraction as an equivalent fraction with the least common denominator.

$$\text{Vegetable Garden: } \frac{2}{3} = \frac{10}{15}$$

$$\text{Fruit Orchard: } \frac{1}{5} = \frac{3}{15}$$

- (c) Add the fractions. Simplify your answer, if possible.

$$\text{Fraction of Land Used for Food} = \frac{13}{15}$$

Problem 7 Factory farms store animal waste in massive open-air pits called lagoons. In one community, a lagoon is $\frac{5}{6}$ full of untreated waste. To lower the level and prevent an overflow into the local groundwater, the farm operators spray an amount equal to $\frac{1}{4}$ of the lagoon's total capacity into the air and onto nearby fields. Follow the steps below to determine the fraction of the lagoon's total capacity that remains full.

- (a) Find the LCD of the fractions.

$$\text{LCD} = 12$$

Hint: To find the LCD, write the multiples of 6 and look for the smallest number that is evenly divisible by 4. Or write the multiples of both numbers and look for the smallest number that appears in both lists.

- (b) Rewrite each fraction as an equivalent fraction with the least common denominator.

$$\text{Initial Waste Level: } \frac{5}{6} = \frac{10}{12}$$

$$\text{Waste Removed: } \frac{1}{4} = \frac{3}{12}$$

- (c) Subtract the fractions. Simplify your answer, if possible.

$$\text{Remaining Waste Level} = \frac{7}{12}$$

Feedback(attempt): While spraying the waste on fields prevents the lagoon from overflowing, it creates a new problem for the community. The waste is often sprayed as a fine mist that can travel for miles, carrying harmful bacteria, nitrates and intense odors into nearby neighborhoods. For the people living nearby, this isn't just an unpleasant smell—it is a serious health hazard! Studies have shown that families living near these spray fields experience higher rates of asthma, respiratory infections and even contaminated well water.